



Human sweet, umami and bitter taste receptors: From decoders of taste perception to sentinels of systemic metabolism and health

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ABSTRACT

Background: Human sweet, umami, and bitter taste receptors, belonging to the GPCR superfamily, are recognized not only as taste mediators but also as crucial chemical sensors regulating systemic homeostasis. However, a lack of high-resolution structural data has long hindered mechanistic understanding and the development of targeted interventions. While recent reviews have touched on their physiological roles, a comprehensive synthesis integrating the latest structural breakthroughs with their expanding systemic functions remains lacking.

Scope of review: This review presents the first comprehensive synthesis of cryo-EM and AI-driven structural breakthroughs in human sweet, umami and bitter taste receptors, detailing their unique ligand recognition modes. It systematically explores the physiological roles of these receptors in taste coding and systemic homeostasis, while evaluating their potential as biomarkers and therapeutic targets for precision nutrition and pharmacological intervention.

Major conclusions: High-resolution structures of TAS2R14, TAS2R43, TAS2R46 and the TAS1R2/TAS1R3 heterodimer have filled critical knowledge gaps, providing a framework for AI-driven structural prediction, ligand screening and rational drug design. Functionally, taste receptors serve as pivotal chemical sensors for dietary components, microbial metabolites and endogenous signals. By initiating highly conserved Ca²⁺ cascades and specific downstream pathways, they regulate taste perception, metabolic homeostasis, immune defense, neurological function and cell fate. Consequently, they significantly influence dietary preferences and are profoundly linked to the susceptibility and prognosis of various diseases, positioning them as promising targets for precision nutrition, disease diagnostics and novel drug development. However, translating this potential into human health applications requires a concerted, cross-disciplinary integration of foundational and translational science.

1. Introduction

As an evolutionarily conserved chemosensory system, taste guides ingestion behavior by initiating immediate hedonic or aversive responses, serving as the body's first line of defense in identifying nutrients and avoiding potential toxins (Hao et al., 2024). Human taste primarily comprises five basic modalities: sweet, umami, bitter, salty and sour. Among these, sweet and umami indicate the presence of energy sources

and nutrients, such as carbohydrates and proteins, respectively, while bitter primarily warns of potential toxic substances. Salty and sour tastes are mainly associated with electrolyte balance and the identification of unripe or spoiled food, respectively (Li et al., 2024). Taste perception is initiated when tastants dissolved in saliva activate specific receptors on taste bud cells. Sweet, umami and bitter tastes are mediated by G protein-coupled receptors (GPCRs) on type II taste cells, whereas salty and sour taste receptors are less extensively characterized and primarily

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involve ion channels on type III cells, such as epithelial sodium channel (ENaC)/transmembrane channel-like protein 4 chloride channel for salt and otopenin-1 for sour, respectively (Roper & Chaudhari, 2017; Zhang et al., 2019). This activation triggers intracellular signaling cascades, ultimately converting chemical signals into neural impulses (Banik & Medler, 2021; Liu, Bayinbata, et al., 2025). Among these taste receptors, GPCRs are particularly pivotal. Sweet (TAS1R2/TAS1R3 heterodimer) and umami (TAS1R1/TAS1R3 heterodimer) receptors belong to the TAS1R family of class C GPCR, and bitter taste receptors consist of 26 TAS2R subtypes belonging to the TAS2R family of class T GPCR. Given the fundamental divergences in structures and signal transduction mechanisms between ion channels and GPCRs, and GPCRs represent highly promising therapeutic targets (Gu et al., 2025), this review will focus exclusively on sweet, umami and bitter taste GPCRs. Specifically, the classification of “fat taste” (i.e., the perception of fatty acids) as a primary basic taste modality remains a subject of ongoing scientific debate. To avoid conceptual ambiguity and maintain a rigorous thematic focus on universally established GPCR-mediated modalities, putative fat receptors fall beyond the scope of this review.

In recent years, with rapid advances in technologies such as high-throughput sequencing and transcriptomics, researchers have gradually discovered the widespread expression of taste receptors in non-gustatory tissues, including the digestive, respiratory, nervous, cardiovascular and reproductive systems (referred to as ectopic taste receptors, in contrast to their traditional expression in the oral cavity) (Rudolph et al., 2025). Although encoded by the same genes as their oral counterparts, ectopic taste receptors, due to their distinct cellular microenvironment and downstream signaling pathways, have expanded their physiological functions beyond taste coding to systemic homeostasis by recognizing dietary components, microbial metabolites and endogenous signaling molecules (Sarnelli et al., 2019; Zhai et al., 2026). Accumulating evidence indicates that taste receptor expression is deeply associated with the susceptibility to and prognosis of obesity, type 2 diabetes, asthma and various cancers (Chou, 2021; Liszt et al., 2022; Yasin et al., 2024). Accordingly, some natural and synthetic compounds targeting taste receptors have entered preclinical or clinical research phases (Yang et al., 2025). Notably, extensive single nucleotide polymorphisms in taste receptor genes contribute not only to inter-individual differences in sensory preference but also serve as a critical nexus linking genetic background, dietary behavior and health outcomes (Yildiz et al., 2016). Despite their growing physiological significance, the inherent low expression levels, hydrophobicity and high conformational dynamics of GPCRs have made it extremely challenging to obtain high-quality protein crystals, leading to a long-standing lack of high-resolution structural data for taste receptors. This bottleneck has severely hindered a deep understanding of their mechanisms in chemosensation and systemic homeostasis, as well as structure-based ligand screening and drug design. Currently, a diverse array of exogenous ligands with rich chemical structures for taste receptors, has been identified from food, the environment and pharmaceuticals. This has not only expanded our understanding of the broad-spectrum chemical recognition properties of taste receptors but also provided a crucial foundation for developing taste modulators (Ahmed et al., 2010; Iwaniak et al., 2016; Yue et al., 2025; Ziaikin et al., 2025). However, the endogenous ligand repertoire for taste receptors remains poorly understood, with some TAS2R subtypes (e.g., TAS2R19, TAS2R45, TAS2R60) remain “orphan receptors”, posing a critical obstacle to comprehensively decoding the functions of taste receptors. Excitingly, with the development of cryo-electron microscopy (cryo-EM) and artificial intelligence (AI) technologies, high-resolution structures of TAS2R46, TAS2R14, TAS2R43 and the TAS1R2/TAS1R3 heterodimer, along with their unique ligand recognition and activation mechanisms have been elucidated (Hu et al., 2024; Juen et al., 2025; Kim et al., 2024, 2026; Shi et al., 2025; Tao et al., 2024; Wang et al., 2025; Xu et al., 2022). These milestone achievements not only profoundly reveal the mechanisms of taste receptors in taste coding and systemic homeostasis but also lay a

solid structural foundation for precisely discovering functional ligands and rationally designing highly selective and potent agonists or allosteric modulators.

Given the interspecies differences in ligand recognition and activation mechanisms, this review focuses on human GPCR-type taste receptors to ensure the most direct translational relevance for understanding human health and disease. This review comprehensively summarizes the high-resolution structural features of the TAS1R and TAS2R families elucidated by cryo-EM, as well as computational strategies for taste receptor structure prediction and ligand recognition. It delves into the systemic physiological functions and mechanisms of taste receptors in taste coding, multisensory interaction and systemic homeostasis. Furthermore, it outlines the potential of taste receptors as critical targets for precision nutritional intervention, disease diagnosis, prognosis assessment and drug development. The overarching aim is to advance our understanding of taste perception and systemic regulatory mechanisms and to provide scientific guidance and directions for further taste receptor research, including deorphanization, ligand screening and design, personalized dietary interventions, novel non-invasive disease diagnosis strategies and the development of taste receptor-targeted therapeutics.

2. Structural basis and ligand recognition of taste receptors

2.1. Basic classification and tissue distribution of taste receptors

In accordance with the five basic taste qualities, taste receptors are primarily classified into sour, sweet, bitter, salty and umami receptors, which are predominantly localized in the taste bud cells of the gustatory papillae epithelium on the palate, tongue, epiglottis, throat and larynx (Fig. 1A) (Roper & Chaudhari, 2017). To date, the 26 identified TAS2R subtypes exhibit a wide range of activation thresholds and display either narrow tuning (recognizing one or a few specific bitter compounds, e.g., TAS2R3/5) or broad tuning (recognizing a wide array of bitter compounds, e.g., TAS2R10/14/46), and are thus referred to as narrowly tuned and broadly tuned bitter receptors, respectively (Meyerhof et al., 2010). Based on whether they are expressed in the oral cavity, taste receptors are classified into oral/general taste receptors and ectopic taste receptors (Rudolph et al., 2025). The expression of ectopic taste receptor is typically low and dynamically regulated by the local microenvironment, posing significant challenges for their detection and functional investigation (Hu et al., 2024; Kim et al., 2024; Shi et al., 2025; Tao et al., 2024; Xu et al., 2022). Although ectopic taste receptors are the same proteins encoded by the same genes as their oral counterparts, their cellular environment (e.g., intestinal enteroendocrine cells vs. oral taste bud cells) and signaling pathways (e.g., activation of α -gustducin triggering intracellular calcium or cAMP changes to drive hormone secretion such as glucagon-like peptide-1 (GLP-1) vs. activation of α -gustducin triggering neurotransmitter ATP release to transmit the chemical signal to taste nerves) differ fundamentally (Banik & Medler, 2021; Liu, Bayinbata, et al., 2025; Wang et al., 2024). Therefore, taste receptors serve physiological roles in addition to taste perception (Sarnelli et al., 2019). This functional diversity significantly expands our understanding of the biological significance of taste receptors.

2.2. Structure of taste receptors

The structures of taste receptors not only reveal the molecular basis of ligand selectivity and signal transduction but also provide a theoretical foundation for the *de novo* design and optimization of taste compounds, offering a structural biology basis for precision nutrition interventions. In recent years, advances in cryo-EM and AI technologies have synergistically propelled the study of taste receptor structures into a new era. This section integrates insights from experimental structural biology and computational modeling to systematically outline the key

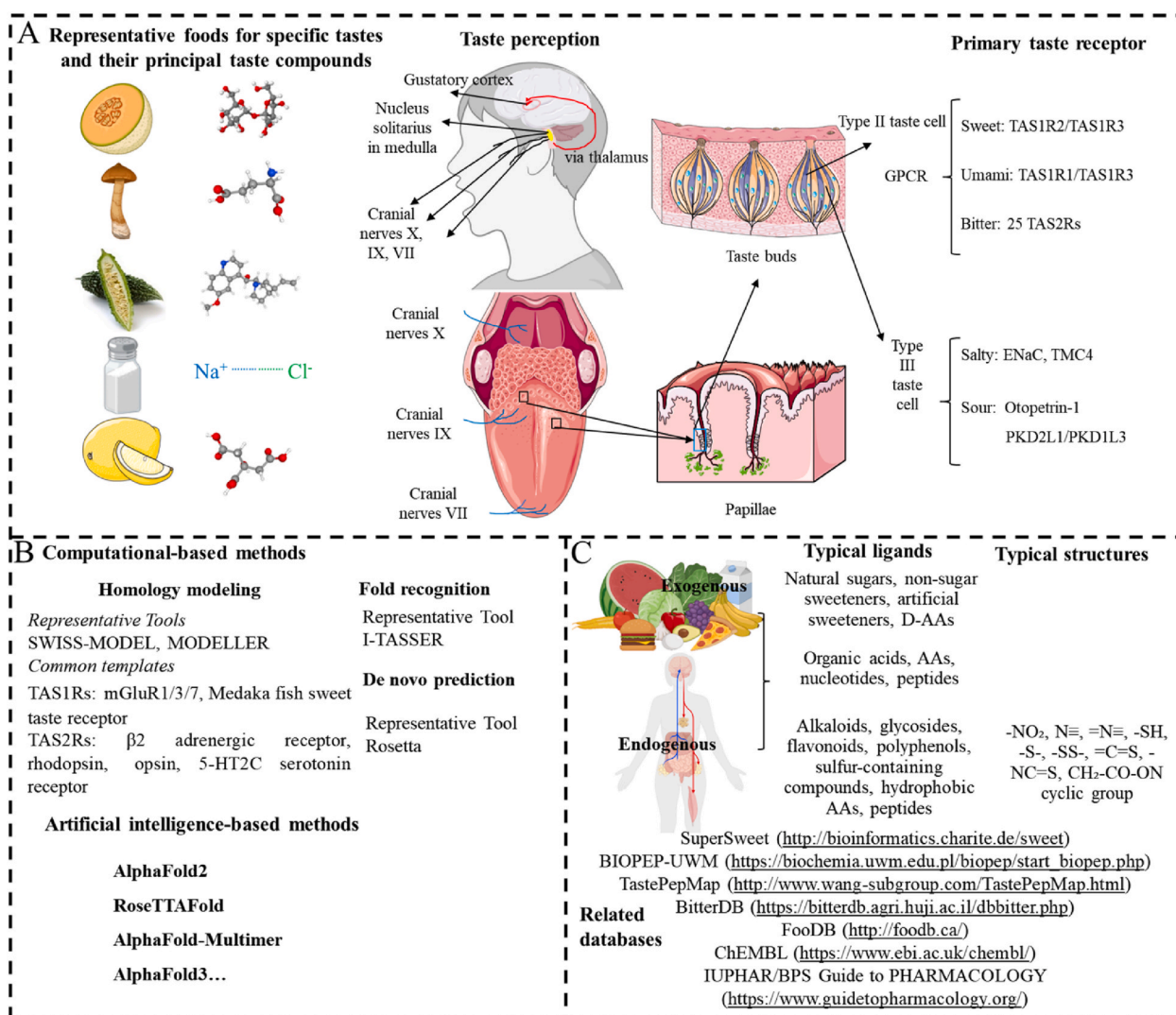


Fig. 1. A comprehensive landscape of human taste receptors. (A) Physiological pathways of taste perception and the classification of taste receptors. (B) Computational methodologies for taste receptor structure prediction. (C) Typical ligands of taste receptors, their typical structural features and associated databases. Figure created using images adapted from [smart.servier.com](https://www.smart.servier.com) and [biorender.com](https://www.biorender.com).

structural features of taste receptors.

2.2.1. High-resolution structural elucidation of taste receptors via Cryo-EM

In 2022, the cryo-EM structures of human TAS2R46 in complex with the chimeric mini-G protein gustducin (PDB ID: 7XP5), in a potent neurotoxin strychnine-bound form (PDB ID: 7XP6) and in the apo form (PDB ID: 7XP4) were reported, representing the first structurally resolved taste receptor and Class T GPCR (Xu et al., 2022). TAS2R46 retains the typical seven-transmembrane (7TM) helical topology of GPCRs but features a broad “baseball cap”-shaped ligand-binding pocket on the extracellular side. Strychnine is stabilized within this pocket through π - π and hydrogen-bond interactions with key residues, such as W88^{3,32} and E265^{7,39}. The study also identified a novel “toggle switch” residue, Y241^{6,48T}, which is crucial for receptor activation and is distinct from classical Class A GPCRs. Interestingly, the apo-state TAS2R46 can also form a complex with G protein, implicating a ligand-independent pre-coupling mechanism.

In 2024, several research groups almost concurrently reported the cryo-EM structures of TAS2R14 in complex with various agonists—including cholesterol, flufenamic acid (FFA), aristolochic acid (AA) and its derivative cmpd28.1—along with downstream G proteins (G_{gust} , associated with bitter perception in taste bud cells; G_{il} , related to

extra-gustatory functions) (PDB IDs: 8VY7, 8VY9, 8XQL, 8XQN, 8XQO, 8XQP, 8XQR, 8XQS, 8XQT, 8YKY, 9IIW, 9IIA, 9IIX, 9IJ9, 8RQL) (Hu et al., 2024; Kim et al., 2024; Tao et al., 2024). These studies yielded consistent and innovative discoveries: (i) Endogenous cholesterol stably occupies the orthosteric pocket (which overlaps with the traditional orthosteric site in most Class A GPCRs) via hydrophobic interactions with the conserved residue W89^{3,32}. Bioluminescence resonance energy transfer and cAMP inhibition functional assays confirmed that cholesterol acts as an orthosteric agonist for TAS2R14. (ii) Highly potent bitter agonists, such as FFA, AA and cmpd28.1, bind to a novel intracellular allosteric pocket jointly formed by the intracellular ends of TM3, TM5, TM6 and TM7. (iii) Agonists bound to this intracellular pocket interact directly not only with the receptor but also with the $\alpha 5$ helix of the G protein, acting as a wedge to directly stabilize the receptor-G protein coupling interface, therefore functioning as dual-role molecules with both agonist and positive allosteric modulator (PAM) properties. Furthermore, some studies also observed a third ligand-binding pocket, further elucidating the molecular basis for TAS2R14's capacity to recognize a vast array of structurally diverse compounds. Collectively, these findings reveal that TAS2R14 integrates and responds to extracellular and intracellular chemical signals through the synergistic action of multiple functionally distinct ligand-binding

sites.

Most encouragingly, the structural understanding of the TAS2R family was further advanced by the recent cryo-EM determination of TAS2R43, a key receptor responsible for perceiving the bitterness of coffee constituents such as caffeine and cafestol (Kim et al., 2026). The cryo-EM structures of human TAS2R43 coupled with either Gi1 (PDB ID: 9OXA) or Ggust (PDB ID: 9OXB) have been resolved. These two structures are nearly identical and exhibit a similar overall 7TM topology to previously reported TAS2R structures. Despite these topological similarities, TAS2R43 displays distinct structural idiosyncrasies. Most notably, it possesses an extended TM5 that reaches further into the extracellular region—a unique feature distinct from previously elucidated TAS2R structures. The structures revealed that AAI binds within a classical extracellular orthosteric pocket of TAS2R43. This binding mode parallels the strychnine recognition in TAS2R46, yet it contrasts sharply with the mechanism of TAS2R14 as previously described. Complementing these cryo-EM findings, long-timescale molecular dynamics simulations identified potential cryptic allosteric pockets within TAS2R43. Collectively, these structural breakthroughs underscore the profound structural diversity and distinct ligand-recognition mechanisms within the TAS2R family, explaining their remarkable capacity to accommodate chemically diverse bitter tastants.

Sweet and umami taste receptors obligatorily function as heterodimers, which poses additional challenges for their structural determination. Shi et al. (2025) successfully resolved the cryo-EM structures of the full-length human TAS1R2/TAS1R3 heterodimer in both the apo state (PDB ID: 9UT8) and the sucralose-bound state (PDB ID: 9UTB). TAS1R2 and TAS1R3 remain constitutively asymmetric, with sucralose binding to the Venus flytrap domain (VFD) of TAS1R2. In the apo state, a 70° angle exists between the VFD (PDB ID: 9UT9) and the transmembrane domain (TMD) (PDB ID: 9UTA) of the dimer, accompanied by a twisted cysteine-rich domain (CRD). The TMD dimerization interface is predominantly formed by TM6 of both subunits and the intracellular parts of TAS1R2-TM5 via hydrophobic and π - π interactions. Sucralose binding triggers three main structural alterations: (i) conformational changes in the dimeric VFDs (PDB ID: 9UTC), characterized primarily by the closure of the TAS1R2 orthosteric pocket and a unidirectional shift of the protruding loop (residues 511–519) in TAS1R3. (ii) CRD bending, primarily resulting from the upward shift of TAS1R2 lobe 2 (LB2), which subsequently affects the C233–C513 disulfide bond connecting LB2 and the CRD. (iii) a slight rotation at the TMD dimerization interface, predominantly occurring within the cytoplasmic regions of TM3, TM7, intracellular loop 2 (ICL2), ICL3 and helix 8 of TAS1R2. In both the apo and bound states, the VFD dimerization interfaces of TAS1R2/TAS1R3 are primarily mediated by LB1 of each subunit, supported by LB2. The intracellular loop 2 (ICL2) in both subunits exhibits a short helical conformation, whereas the extracellular loop 2 (ECL2) adopts a conserved parallel β -sheet conformation, binding tightly in a subunit-specific manner and stacking against the short β -strand at the C-terminus of the CRD. A specific loop in the TAS1R3 CRD (residues 511–519, wherein F514 and H515 are critical for sucralose-induced receptor activation) inserts into the interface between LB2 of each subunit. Additionally, the oblique extension ECL2 of TAS1R3 interacts with residues 542–546 of its CRD, while the ECL2 of TAS1R2 forms hydrophobic and polar interactions with ECL3, enhancing the flexibility of its CRD. Disulfide bonds bridge the VFD and the CRD, as well as ECL2 and TM3. Combining site-directed mutagenesis and molecular dynamics simulations, it was confirmed that sucralose is anchored within the inter-lobe cleft of the TAS1R2-VFD primarily through polar interactions, hydrogen bonds and hydrophobic contacts with key residues including I67, L71, Y103, D142, N143, S165, P277, D278, E302 and V384. Around the same time, Juen et al. (2025) reported a consistent conformation of the sweet receptor-sucralose complex (PDB ID: 9NOR) and identified comparable key binding residues (PDB: 9NOV), further resolving its coupling interface with G protein (miniGs/gust) and revealing that aspartame shares the same binding

pocket in TAS1R2 as sucralose. Furthermore, Wang et al. (2025) successfully captured a novel “loose” intermediate activation state of the sweet receptor upon binding sucralose and advantame, characterized by pronounced separation of the VFDs of the two subunits, with the VFD of TAS1R2 exhibiting maximal closure.

Currently, the structural understanding of the TAS1R1/TAS1R3 heterodimer remains incomplete. However, the successful structural determination of the sweet taste receptor provides a critical framework for the umami receptor, as both are class C GPCR heterodimers sharing the TAS1R3 subunit. Similarly, the structures of TAS2R14, TAS2R43 and TAS2R46 serve as valuable references for structural studies of other bitter taste receptors.

2.2.2. Computational and AI-driven structural prediction of taste receptors

In the context of a general lack of high-resolution structures for taste receptors, computational (e.g., homology modeling, fold recognition and *de novo* prediction) and AI-based approaches have become critical tools for their structural prediction (Fig. 1B). The theoretical foundation of homology modeling is predicated on the principle that evolutionarily related proteins tend to conserve their three-dimensional structures. This computational approach constructs a model by exploiting the sequence homology between a target protein and a known structural template, with SWISS-MODEL and MODELLER being among the most widely utilized tools for this purpose. When modeling TAS1R1, researchers typically employ metabotropic glutamate receptors (e.g., mGluR1, mGluR3 and mGluR7), alongside more phylogenetically proximate taste receptors (e.g., the extracellular domain of the Medaka fish sweet taste receptor (PDB ID: 5 × 2M)) as templates to construct the critical VFD (Spaggiari et al., 2020). For bitter taste receptors, structural models are predominantly built utilizing the β 2-adrenergic receptor, rhodopsin (PDB ID: 1U19), opsin (PDB ID: 3DQB) and the 5-HT2C serotonin receptor (PDB ID: 6BQC) as templates (Spaggiari et al., 2020). Although homology modeling is widely utilized, its predictive accuracy is heavily contingent upon template quality and sequence similarity. Due to the low sequence identity between taste receptors and available GPCR templates (Di Pizio et al., 2016; Spaggiari et al., 2020), traditional homology models failed to capture highly divergent, receptor-specific topological features (e.g., the intracellular allosteric pocket of TAS2R14 or the extended TM5 of TAS2R43). Because classical templates lack these crucial binding interfaces, early structure-based drug design yielded general predictive validity. Consequently, homology modeling can often be insufficient for elucidating taste receptor function and developing the highly selective pharmacological tools required for rigorous physiological testing. In cases of low sequence similarity, fold recognition methods provide a viable alternative. This approach identifies the most probable folding pattern by evaluating the compatibility of the target sequence with libraries of known structural folds, with I-TASSER being a representative tool. Conversely, in the complete absence of suitable templates, *de novo* modeling relies exclusively on physicochemical principles (e.g., energy minimization) and amino acid sequence information to predict structures, with Rosetta being a representative tool (Alford et al., 2017). While these methods have been applied to the structural modeling of both TAS1Rs and TAS2Rs, the vast conformational space of GPCRs and the imperfections in energy functions make *de novo* prediction of large membrane proteins like taste receptors exceptionally challenging.

Recently, AI has brought a revolutionary breakthrough to protein structure prediction. AlphaFold2 and RoseTTAFold employ deep learning to mine co-evolutionary information from multiple sequence alignments, utilizing the Evoformer and a three-track network architecture, respectively, to predict protein monomer structures with high precision. The quality of many predicted GPCR structures approaches experimental resolution. AlphaFold2 has already been utilized to predict the full-length structures of all human umami, sweet and bitter taste GPCRs (Fleming et al., 2025). Concurrently, the rapid iteration of AI tools has led to updated versions like AlphaFold-Multimer, which is

capable of directly predicting the TAS1R2/TAS1R3 heterodimer and its interactions with ligands. Furthermore, the release of AlphaFold3 in May 2024 marked a pivotal breakthrough, expanding its predictive capabilities beyond proteins to encompass virtually all biological macromolecules and their complexes with ligands. Shimizu et al. (2025) confirmed that AlphaFold3 yields more accurate structural predictions for human TAS2Rs compared to its predecessor, AlphaFold2; this model is now accessible to the academic community via the AlphaFold Server. Despite the robust predictive power of AI methods, they primarily predict the most probable static conformations and struggle to directly capture the dynamic conformational changes that occur during receptor activation. Consequently, integrating high-precision AI models with molecular dynamics simulations has emerged as a predominant research strategy for investigating the dynamic functionalities, allosteric modulation mechanisms and drug discovery avenues of taste receptors. Ultimately, *in silico* predictions necessitate further rigorous experimental validation. For instance, critical residues implicated in ligand binding or signal transduction, as identified by the structural models, should be corroborated through site-directed mutagenesis and functional assays. Concordance between experimental mutagenesis data and computational predictions provides compelling validation for the hypothesized binding pockets, thereby confirming the structural integrity of the model and its physiological relevance.

It is particularly emphasized that the convergence of high-resolution structural biology and AI has shifted the taste receptor field from descriptive phenomenological observations to mechanism-driven research. No longer are we limited to merely correlating receptor expression with physiological outcomes; instead, we can now rationally answer how specific ligands interact with receptors to initiate the signaling cascades that exert gustatory or other physiological functions. Therefore, understanding the structural basis of taste receptors—such as their unique binding pockets—is the key prerequisite for advancing the development of precision nutritional strategies and targeted therapeutics detailed in the following sections.

2.3. Exogenous and endogenous ligands of taste receptors: identification methods, chemical diversity and databases

Taste receptors mediate dietary behavior and digestive responses by recognizing exogenous ligands from the food and environment, while simultaneously monitoring and modulating immune and metabolic processes through the detection of endogenous ligands. Current research predominantly focuses on exogenous ligands, whereas their endogenous counterparts remain poorly characterized. Comprehensively summarizing the key ligand identification methods and repertoires of taste receptors is vital for understanding systemic chemosensing, achieving precision nutrition and developing targeted drugs. Beyond sensory evaluations, ligand (e.g., agonists, antagonists) identification also relies on cellular models heterologously expressing specific taste receptors by monitoring calcium mobilization via fluorescent probes (Belloir et al., 2021). Furthermore, the high-resolution determination and high-precision predictions of taste receptor structures, coupled with computational approaches such as molecular docking and molecular dynamics simulations, have significantly advanced ligand screening, identification and design. Moreover, AI technologies (e.g., quantitative structure-activity relationship analysis, pharmacophore modeling and machine learning) facilitate ligand screening by constructing predictive models based on common chemical features of known ligands. Beyond that, they enable the *de novo* design of novel taste-active molecules, thereby advancing ligand research from the stage of “discovery” toward “creation”. For ectopic taste receptors, endogenous ligands can be efficiently identified by coupling *in vitro* functional assays with metabolomics to profile dynamic metabolite fluctuations in the local microenvironment under specific physiological (e.g., postprandial) or pathological (e.g., bacterial infection) conditions (Nagasato et al., 2025).

Studies have revealed that the TAS1R2/TAS1R3 heterodimer employs a multi-site recognition mechanism to detect a diverse array of ligands, including natural sugars, non-sugar sweeteners (e.g., thaumatin, monellin, brazzein, steviol glycosides), artificial sweeteners (e.g., saccharin, sucralose, acesulfame potassium, neotame) and D-amino acids (Fig. 1C) (Wang et al., 2025). Structurally, this broad tuning is facilitated by multiple binding pockets within the receptors. While many studies indicate that sugars predominantly bind to the TAS1R2-VFD (Hao et al., 2024), the sugar-bound state has not yet been fully elucidated at the molecular structural level. However, recently resolved structures have demonstrated that sweeteners like sucralose and aspartame bind to overlapping binding sites within the orthosteric site of the TAS1R2-VFD (Juen et al., 2025; Shi et al., 2025). Additionally, certain PAMs, such as SE-1, SE-2 and SE-3, also interact within the TAS1R2-VFD. In contrast, the TMD, particularly that of TAS1R3, serves as a critical interaction site for certain negative allosteric modulators like lactisole (Jiang et al., 2005). Common umami tastants include organic acids, amino acids (e.g., L-glutamate), nucleotides (e.g., IMP and GMP) and peptides. Specifically, L-glutamate binds to the orthosteric site of the TAS1R1 VFD, while IMP and GMP act as potent PAMs by binding to an allosteric site near the VFD entrance. This concurrent binding stabilizes the closed conformation of the VFD, thereby generating a synergistic umami-enhancing effect (Belloir et al., 2025). TAS2Rs achieve broad-spectrum recognition via combinatorial coding; a specific ligand can activate multiple TAS2R subtypes, and conversely, a single TAS2R can often respond to an array of structurally diverse ligands. While the ligands for TAS219, TAS245 and TAS260 remain uncharacterized, known ligands for other TAS2Rs exhibit extensive structural diversity, typically featuring bitter-determining motifs—such as -NO_2 , $\text{N}\equiv\text{N}$, =N , -SH , -S- , -SS- , =C=S , -NC=S and $\text{CH}_2\text{-CO-ON}$ cyclic groups (Li et al., 2024)—encompassing alkaloids (e.g., quinine, caffeine, strychnine and nicotine), glycosides (e.g., salicin and soyasaponins), flavonoids and polyphenols (e.g., naringin, catechins and quercetin), sulfur-containing compounds (e.g., glucosinolates and isothiocyanates), hydrophobic amino acids and peptides (Tian et al., 2025). TAS1Rs and TAS2Rs, expressed in extraoral tissues such as enteroendocrine cells, respond to endogenous molecules including luminal nutrients (e.g., sugars and amino acids), microbiota-derived metabolites such as short chain fatty acids (which regulate the expression of TAS1Rs; Shackley et al., 2020) and acyl-homoserine lactones (AHLs), (which directly activate TAS2R38; Lee et al., 2012), primary, secondary and tertiary bile acids, cholesterol and steroid hormones (which specifically activate TAS2R14 and TAS2R46; Lang et al., 2026). This chemosensory activation triggers the release of gut hormones, thereby regulating systemic energy metabolism (Sarnelli et al., 2019). Furthermore, TAS2Rs expressed in airway epithelial cells, such as TAS2R38, recognize quorum-sensing molecules produced by *Pseudomonas aeruginosa* (e.g., acyl-homoserine lactones). Upon activation, these receptors mediate pathogen clearance via mechanisms including nitric oxide (NO) production and ciliary beating.

Currently, extensive chemosensory data have been integrated into specialized databases. These include BitterDB (<https://bitterdb.agri.huji.ac.il/dbbitter.php>) for bitter receptors and their ligands, the Super-Sweet database (<http://bioinformatics.charite.de/sweet>) for sweet molecules and the BIOPEP-UWM (https://biochemia.uwm.edu.pl/biopep/start_biopep.php) and TastePepMap (<http://www.wang-subgroup.com/TastePepMap.html>) databases for taste peptides (Ahmed et al., 2010; Iwaniak et al., 2016; Yue et al., 2025; Ziaikin et al., 2025). Furthermore, broader resources such as FooDB (<http://foodb.ca/>) for food constituents, ChEMBL (<https://www.ebi.ac.uk/chembl/>) for drug and drug-like molecules (Gaulton et al., 2017), and the IUPHAR/BPS Guide to PHARMACOLOGY (<https://www.guidetopharmacology.org/>)—an expert-driven knowledgebase of drug targets and their ligands—provide invaluable complementary data (Harding et al., 2023). Querying for specific taste modalities or receptor targets on these platforms will retrieve the relevant tastant compounds or

corresponding ligands.

3. Systems biology of taste receptors: From oral sentinels to systemic homeostasis regulators

3.1. Regulation of taste receptors in taste coding

The perception of sweet, umami and bitter tastes originate from the specific taste receptors, which selectively recognize tastants dissolved in saliva. This chemical stimulation is then transduced into neural electrical signals via a highly conserved intracellular signaling cascade, thereby forming taste perception (Fig. 2) (Banik & Medler, 2021; Liu, Bayinbata, et al., 2025). This process not only elicits immediate behavioral feedback, such as hedonic or aversive responses to food, but also maintains energy homeostasis and acts as the body's first line of defense against exogenous toxic substances (Hao, Fan, et al., 2024). Upon receptor activation by specific tastants, the coupled heterotrimeric G protein dissociates, releasing $G\alpha$ and $G\beta\gamma$ subunits. The $G\beta\gamma$ subunit activates phospholipase $C\beta 2$ (PLC $\beta 2$), which catalyzes the hydrolysis of PIP $_2$ into diacylglycerol and inositol 1,4,5-trisphosphate (IP $_3$). IP $_3$ binds to IP $_3$ receptors (IP $_3$ R) on the endoplasmic reticulum, triggering Ca^{2+} release from intracellular stores. This elevated intracellular Ca^{2+} subsequently activates the membrane transient receptor potential channels (e.g., TRPM4 and TRPM5), leading to Na^+ influx and cellular depolarization. This depolarization drives the release of the neurotransmitter ATP into the synaptic cleft via voltage-gated calcium homeostasis modulator1/3 (CALHM1/3) channels (Fig. 2). Furthermore, during sweet, umami and bitter taste perception, $G\alpha$ -gustducin regulates phosphodiesterase activity and decreases cAMP levels, which relieves the cyclic nucleotide-mediated inhibition of ion channels, facilitating Ca^{2+} influx and depolarization (Li et al., 2024; Liu, Bayinbata, et al., 2025). The cell-specific expression of these receptors, coupled with the corresponding labeled-line neural transmission pathways, underlies the specific coding of taste modalities (Diepeveen et al., 2022). Upon release from activated taste cells, ATP specifically acts on P2X $_2$ /P2X $_3$ receptors on primary sensory nerve terminals that form stable synapses with these cells, transducing the chemical signal into

action potentials. Although action potentials are uniform in nature, sweet, umami and bitter propagation along specific fibers functions as coded information transmitted via dedicated lines. These signals are transmitted through the facial (CN VII), glossopharyngeal (CN IX) and vagus (CN X) nerves to the nucleus of the solitary tract in the brainstem. They are subsequently relayed through the ventral posteromedial nucleus of the thalamus parvocellular part and projected to higher-order gustatory processing areas, such as the insular cortex (Li et al., 2024). The brain ultimately decodes the taste experience by identifying the source of the signal along these “labeled lines” (i.e., distinguishing which specific cluster of fibers has been activated). This complete chain—from peripheral receptor recognition and ubiquitous intracellular signal transduction to central decoding via dedicated pathways—constitutes the biological foundation of taste perception.

3.2. Interactive regulation of taste receptors in multisensory integration

Taste receptors are not only precisely regulated by the physical environment of the oral cavity but are also deeply involved in cross-modal sensory integration. A notable thermo-gustatory interaction exists, for instance, warmth can enhance sweet perception, whereas tea tastes more bitter upon cooling (Talavera et al., 2005; Xie et al., 2024). The former may be attributed to the temperature sensitivity of TRPM5 (Uchida, 2024), whose open probability increases with elevated temperatures, thereby amplifying the depolarization currents mediated by TAS1R2/TAS1R3. Concurrently, food texture (e.g., viscosity) alters the diffusion rate of tastant molecules to receptor, regulating the effective ligand concentration and consequently modulating taste intensity (Wu & Zhao, 2021). Furthermore, the activation of taste receptors sends ingestive signals to the central nervous system, serving as a primary factor in triggering the oral referral of retronasal odors—a perceptual process where retronasal olfactory inputs are perceived as originating from the oral cavity due to their congruency with concurrent gustatory inputs (Lim & Johnson, 2012). This enhances the sensitivity of the olfactory cortex to retronasal odors and spatiotemporally binds olfactory and gustatory signals within the orbitofrontal cortex, forming a flavor experience with spatial localization and tangibility (Rolls, 2012). In the

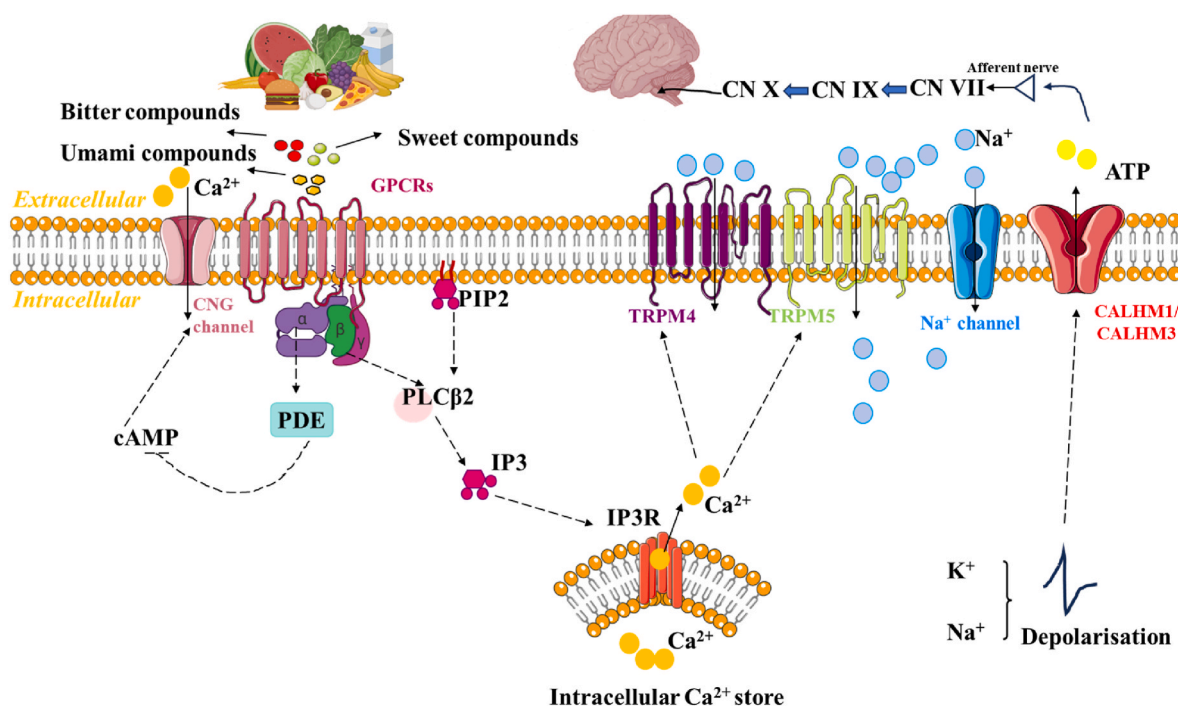


Fig. 2. Intracellular signaling cascades and neural transmission pathways underlying human sweet, umami and bitter taste perception. Figure created using images adapted from [smart.servier.com](https://www.smart.servier.com) and [biorender.com](https://www.biorender.com).

absence of gustatory inputs, pure olfactory stimuli often fail to elicit a complete sense of food reward. Such cross-modal crosstalk ensures that the organism forms a multidimensional and precise perception of food.

3.3. Beyond sensation: Systemic regulation of whole-body homeostasis by ectopic taste receptors

In recent years, facilitated by cutting-edge interdisciplinary technologies—such as high-throughput sequencing and molecular biology for identifying gene and protein expression, along with optogenetics and electrophysiology for validating functional signaling—studies have revealed the widespread distribution and physiological significance of taste receptors across various non-gustatory tissues. As chemosensors distributed throughout the body, ectopic taste receptors monitor fluctuations in nutrients, metabolites and microbiota within the cellular microenvironment, exerting vital physiological functions. This section will systematically elucidate the potential physiological roles of ectopic taste receptors beyond gustatory perception.

3.3.1. Metabolic regulation and energy homeostasis

As the largest interface for host-environment interaction, taste receptors (TAS1R1/2/3 and a wide array of TAS2Rs, including TAS2R1/3/4/5/7/8/10/13/14/16/20/30/31/38/39/40/42/43/45/46/50/60) expressed in the gastrointestinal mucosa continuously monitor the

intraluminal nutritional state to coordinate systemic energy homeostasis (Sternini & Rozengurt, 2025; Zhang et al., 2026). The activation of gastric TAS2Rs mediates gastric acid secretion in parietal cells and inhibits gastric emptying via human gastrointestinal smooth muscle cells (Chen et al., 2025). Upon activation by bile acids, denatonium benzoate and polyphenols, intestines TAS2Rs initiate a Ca^{2+} signaling cascade highly conserved with that in taste buds, regulating host metabolism primarily through three distinct pathways (Fig. 3A): (i) Endocrine regulation. Activated TAS2R1/4/5/9/14/38/39/43/46 promote the release of anorexigenic hormones (e.g., GLP-1, peptide YY and cholecystokinin) from enteroendocrine cells (EECs), while suppressing the secretion of orexigenic hormones like ghrelin (Wang et al., 2024). These hormones act via endocrine routes on the pancreas and modulate gastrointestinal motility (e.g., delaying gastric emptying) to reduce blood glucose and energy intake. (ii) Neural transmission. GLP-1 and peptide YY can also directly activate the enteric nervous system and the vagus nerve, mediating gut-brain signaling. Furthermore, ATP and 5-Hydroxytryptamine released by intestinal cells serve as paracrine signals to enhance vagal activation. Certain EECs even form synaptic connections with the vagus nerve via “neuropods” rapidly transmitting signals (in milliseconds) to the central nervous system to suppress feeding behaviors (Barton et al., 2023). (iii) Immune defense. Bitter compounds stimulate jejunal crypts (particularly Paneth cells) to release antimicrobial peptides, fortifying intestinal barrier function, inhibiting bacterial translocation and ameliorating inflammation-associated insulin

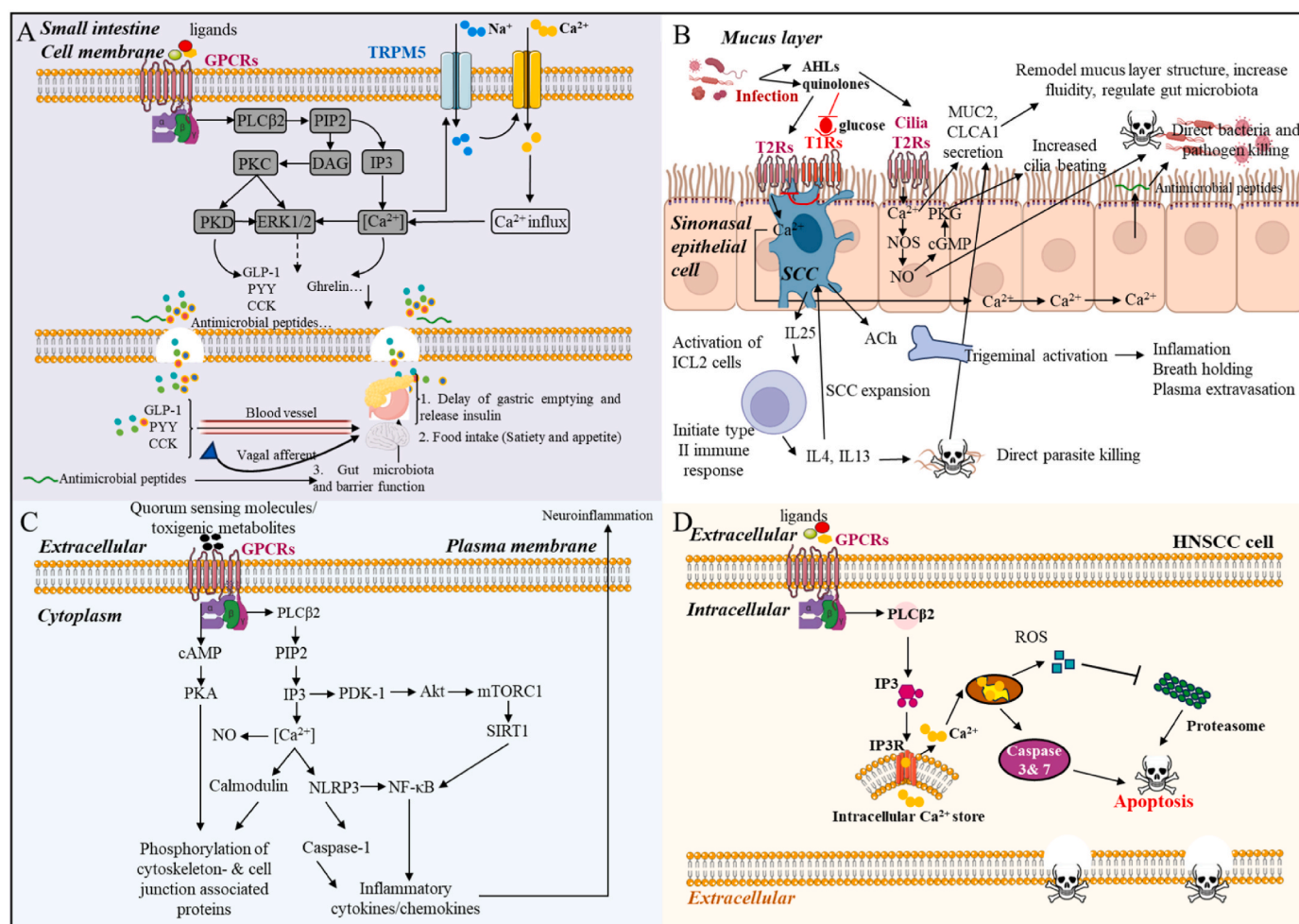


Fig. 3. Ectopic functions and diverse signaling mechanisms of taste receptors in extra-oral tissues (Avau & Depoortere, 2016; Carey & Lee, 2019; Miller et al., 2025; Raka et al., 2019; Sternini & Rozengurt, 2025; Welcome & Mastorakis, 2021). (A) Metabolic regulation and energy homeostasis in the small intestine. (B) Innate immune defense and barrier protection in sinonasal epithelial cells. (C) Neuroprotection and anti-inflammatory signaling. (D) Induction of apoptosis in head and neck squamous cell carcinoma cells. Figure created using images adapted from [smart.servier.com](https://www.smart.servier.com) and [biorender.com](https://www.biorender.com).

resistance and metabolic dysfunction (Tong et al., 2025). Similarly, certain animal and pre-clinical studies revealed that activation of sweet taste receptors not only trigger gut hormone release but also upregulates sodium-glucose linked transporter 1 (SGLT-1) expression to directly enhance glucose absorption (Su Khin et al., 2026). Furthermore, sweet receptors differentially transmit caloric signals to the brain via distinct neurotransmitter pathways: non-nutritive sweeteners predominantly induce purinergic (e.g., ATP) transmission, whereas sugars such as glucose activate SGLT-1-dependent glutamatergic transmission (Buchanan et al., 2022).

3.3.2. Immune defense and barrier protection

Taste receptors are extensively distributed across mucosal barriers (e.g., respiratory tract, gastrointestinal tract, oral cavity and skin) and various immune cells. By specifically recognizing bacterial quorum-sensing molecules and toxins, they rapidly initiate multi-dimensional defensive responses encompassing physical clearance, chemical killing and cellular immunity. Ectopic taste receptors not only regulate energy homeostasis but also monitor microbial signals and modulate the immune microenvironment (Fig. 3B). Firstly, bitter taste receptors mediate innate immune defense against bacteria in a subtype-specific manner. For instance, TAS2R10 (activated by denatonium benzoate or cucurbitacin E) and TAS2R43 (activated by aloin) induce Paneth cells to release antimicrobial peptides such as α -defensins (e.g., DEFA5, DEFA6), lysozyme and REG3A, directly inhibiting the proliferation of pathogenic bacteria like *E. coli* (Liszt et al., 2022). Notably, antimicrobial peptide release is also ligand-dependent; for example, CuE selectively induces DEFA5 expression. Simultaneously, TAS2R43 can additionally stimulate goblet cells to secrete mucin 2 and chloride channel accessory 1 (CLCA1), remodeling mucus layer structure, increasing fluid mobility and modulating microbiota distribution (Liszt et al., 2022). Secondly, TAS1Rs and their downstream α -gustducin signaling axis, enriched in tuft cells (also known as brush cells or solitary chemosensory cells), constitute a specific defense line against parasitic infections. As demonstrated by Gerbe et al. (2016), this mechanism involves releasing IL-25 to activate group 2 innate lymphoid cells, initiating a type 2 immune response that promotes goblet cell hyperplasia and mucus secretion. This chemo-immune function exhibits pathophysiological plasticity and genetic dependence. For example, jejunal crypts in obese patients display heightened sensitivity to bitter agonists, leading to significantly augmented TAS2R10-mediated antimicrobial peptide release (e.g., β -defensin 1 is triggered exclusively under obese pathological conditions). This may represent a compensatory mechanism against dysbiosis and barrier damage, sequestering high-fat diet-induced microbial imbalance and endotoxins to prevent their systemic translocation and subsequent chronic inflammation. Furthermore, host genetic backgrounds (such as TAS2R43 gene deletion polymorphisms) can ablate specific ligand-induced immune responses (e.g., aloin-induced CLCA1 release) (Liszt et al., 2022), indicating that metabolic and genetic contexts collectively influence intestinal immune efficacy.

Similarly, as the primary barrier against environmental insults, the skin relies on taste receptors to maintain its immune homeostasis. Upon exposure to harmful bitter substances, keratinocytes utilize the functional coupling between TAS2R signaling and ATP-binding cassette transporter B1 to facilitate the efflux of harmful compounds, thereby enhancing skin resistance to damage (Mori et al., 2024). Specifically, TAS2R38 is expressed not only in skin structural cells but also on the surface of skin-infiltrating CD4⁺ and CD8⁺ T cells, as well as peripheral blood mononuclear cells. Activation of TAS2R38 by phenylthiocarbamide and 6-n-propylthiouracil significantly inhibits the chemotactic migration of immune cells, suggesting that TAS2R38 may act as a “brake” to restrict excessive immune cell infiltration to inflammatory sites and prevent immune-mediated tissue damage (Sakakibara et al., 2022). In lesional skin of atopic dermatitis patients, TAS2R38 expression is upregulated and positively correlates with disease severity markers, such as serum IgE, thymus and activation-regulated

chemokine, indicating that aberrant TAS2R38 may be associated with the excessive recruitment of immune cells under inflammatory conditions. Interestingly, endogenous substances capable of activating TAS2R38 have been detected in the skin, implying the existence of an endogenous ligand system targeting this receptor in the skin. Further identification of these key endogenous ligands will provide vital targets for the development of novel anti-inflammatory therapeutics.

In the nasal cavity and paranasal sinus epithelium, ciliated epithelial cells and solitary chemosensory cells also widely express taste receptors such as TAS2R4/14/16/38/46. When receptors like TAS2R38, localized on motile cilia, are activated by bacterial quorum-sensing molecules (e.g., AHLs produced by *Pseudomonas aeruginosa*, competence-stimulating peptides (CSPs) from gram-positive bacteria and quinolone derivatives), they initiate a nitric oxide synthase-dependent signaling pathway. This generates NO and activates protein kinase G, which increases ciliary beat frequency, accelerating the physical clearance of the mucus layer and disrupting bacterial biofilm formation (Chen et al., 2025). Concurrently, squamous cells interspersed in the epithelium act as “alarms”. Upon activation, they release acetylcholine to activate adjacent trigeminal nerve fibers, triggering protective reflexes like sneezing, and employ axonal reflexes to stimulate surrounding epithelial cells to secrete antimicrobial peptides such as β -defensins, thereby constructing a dual defense line of physical clearance and chemical eradication (Xi et al., 2022). Notably, sweet and umami receptors in the respiratory epithelium can modulate bitter receptor-mediated immune responses to prevent immune overactivation. In the lower respiratory tract, TAS2R10/14/31, which are highly expressed in airway smooth muscle, serve as critical hubs for regulating airway tone (Zhang et al., 2025). Studies demonstrate that TAS2R agonist-induced bronchodilation is superior to that of conventional β 2-adrenergic receptor agonists (e.g., isoproterenol) and is less prone to desensitization in inflammatory environments (Tong et al., 2025). The underlying relaxation mechanism involves multiple pathways: (i) Receptor activation triggers localized compartmentalized release of intracellular Ca²⁺, specifically opening large-conductance calcium-activated potassium channels, leading to cell membrane hyperpolarization and muscle relaxation (Deshpande et al., 2010). (ii) G α subunits competitively block acetylcholine receptor-coupled contractile signals (Zhou et al., 2022). (iii) TAS2Rs also relieve LIM kinase-mediated inhibition of cofilin via the “PLC β -Par3-Cofilin” axis, and activated cofilin severs the actin cytoskeleton, resulting in smooth muscle relaxation (Woo et al., 2023). Beyond regulating acute airway tone, TAS2R agonists can dose-dependently suppress the hyperproliferation of ASM cells in asthmatic patients by blocking the PI3K/Akt pathway, highlighting their therapeutic potential against chronic airway remodeling (Sharma et al., 2016). Furthermore, TAS2Rs on the surface of pulmonary macrophages and monocytes participate in immune microenvironment regulation. Bitter agonists (e.g., quinine) can enhance macrophage phagocytosis while inhibiting the release of cytokines such as TNF- α , IL-6, CCL3 and CXCL8. Importantly, these functions are genetically dependent. For instance, the PAV haplotype of TAS2R38 (associated with taste sensitivity) can effectively recognize bacterial AHLs and initiate a potent antibacterial response, lowering gram-negative infection risk. Conversely, the AVI haplotype (associated with taste insensitivity) is linked to higher susceptibility to chronic rhinosinusitis and upper respiratory infections (Lee et al., 2012).

Simultaneously, taste receptors (TAS1R1 and TAS2R3/4/9/10/13/14/30/31/38/39/40/41/42/43/45/46/50/60) are also widely expressed in gingival tissues. TAS2R14 in gingival epithelial cells recognizes CSPs secreted by gram-positive bacteria (e.g., *Staphylococcus aureus*) and upregulates the expression of IL-6, IL-8, TNF- α and human β -defensin-2 to directly inhibit bacterial growth (Medapati et al., 2021). Meanwhile, TAS2R38 primarily recognizes AHLs produced by gram-negative bacteria, rapidly initiating innate immune responses to maintain periodontal microecological balance (Xi et al., 2022). Furthermore, receptor genetic polymorphisms are closely associated

with oral health: variations in the sweet taste receptor TAS1R2 (e.g., Ile191Val) can influence caries susceptibility by altering sugar consumption preferences, while the TAS2R38 genotype directly impacts immune defense capabilities and caries risk by modulating the secretion levels of immune factors (e.g., IL-1 α , IL-8) under pathogenic stimulation (Yildiz et al., 2016).

3.3.3. Neural and brain function regulation

Taste receptors are broadly expressed in the hypothalamus, hippocampus, cortex and spinal cord, where they participate in regulating neuronal excitability and maintaining blood-brain barrier (BBB) defense. Studies indicate that TAS1R3 is one of the amyloid-beta ($A\beta$) receptors. $A\beta$ oligomers or elevated L-glutamine in the cerebrospinal fluid of patients with Alzheimer's disease (AD), can activate TAS1R3. This activation promotes the insertion of voltage-gated sodium channels (VGSCs) into the cell membrane via a microtubule-dependent mechanism, contributing to neuronal hyperexcitability in early AD. Conversely, a recent study utilizing whole-cell patch-clamp recordings in Nav1.5-expressing cells demonstrated that the specific activation of TAS1R2 by artificial sweeteners such as saccharin (4 mM) can inhibit VGSC activity (Saito et al., 2025). Targeted modulation of TAS1Rs signaling may offer novel therapeutic strategies to counteract excitotoxicity in neurodegenerative diseases. However, the potential neurological risks associated with artificial sweeteners warrant caution. Long time consumption of artificial sweeteners, such as aspartame and sucralose, can induce oxidative stress, neuroinflammation and BBB disruption within the central nervous system, accompanied by abnormal cerebral blood flow. Furthermore, they may perturb the gut microbiota and their metabolites, impairing the gut-brain axis and accelerating cognitive decline. These neurotoxic effects are particularly pronounced in patients with type 2 diabetes and obesity, where metabolic disturbances and sweetener-induced pathologies may synergistically increase the risk of cognitive impairment and neurodegenerative diseases (Choudhary, 2025).

TAS2Rs in the central nervous system primarily exert protective roles and exhibit region-specific expression patterns. For instance, TAS2R14 is most abundantly expressed in the cerebellum, suggesting its potential involvement in motor control or cognitive integration (Sternini & Rozengurt, 2025). TAS2Rs expressed on BBB pericytes can be activated by specific bitter molecules (e.g., toxic metabolites) in circulation. This activation enhances tight junction integrity and reduces barrier permeability, thereby restricting the entry of neurotoxins into the brain parenchyma (Duarte et al., 2025). Furthermore, TAS2R14 has also been found to regulate the transport of resveratrol across the human blood-cerebrospinal fluid barrier, offering therapeutic promise for brain disorders (Duarte, Rosado, et al., 2020). Notably, the expression of specific TAS2R genes, such as TAS2R14 and TAS2R5, is downregulated in the early stages of AD and Parkinson's disease, with expression levels progressively declining as the disease advances. This suggests that the loss of these receptors may impair the brain's endogenous neuroprotective capacity (Alves et al., 2023). Specific TAS2R agonists can significantly inhibit the NLRP3 and NF- κ B pathways, reduce the release of pro-inflammatory cytokines such as IL-1 β and TNF- α , and subsequently mitigate neuroinflammation, oxidative stress and apoptosis (Fig. 3C) (Tong et al., 2025). Bitter compounds can also activate gastrointestinal TAS2Rs to regulate neuronal function and ameliorate memory impairment via the gut-brain axis and the noradrenergic system (Wu et al., 2025). The neuroprotective effects of bitter compounds have been comprehensively reviewed by (Duarte et al., 2022). Despite the significant therapeutic potential of TAS2R agonists (e.g., cannabinoids, phenolics) for neuroinflammation, challenges remain, including low bioavailability, unclear mechanisms of brain penetration across the BBB and potential side effects like increased gastric acid secretion, akin to classic AD drugs like donepezil. Future studies should elucidate their brain entry mechanisms and develop brain-targeted delivery systems to enhance central efficacy while minimizing peripheral side effects,

thereby fully exerting the therapeutic potential of bitter receptors in neurodegenerative diseases (Wu et al., 2025).

3.3.4. Cell proliferation, apoptosis and differentiation

TAS1Rs primarily function as nutrient sensors and serve as core drivers of cell growth and differentiation. For instance, intestinal TAS1R1/TAS1R3 senses nutritional signals and promotes epithelial cell renewal and repair by activating the mTORC1 pathway (Wauson et al., 2012). However, these TAS1R-mediated growth-promoting mechanisms are frequently hijacked by neoplastic cells to fuel malignant proliferation (Skapinker et al., 2024). In contrast, TAS2Rs generally act as defensive sensors that mediate cell cycle arrest or apoptosis (Wei et al., 2026). A specific expression profile of TAS2Rs (e.g., TAS2R7, TAS2R9, TAS2R10, TAS2R14, TAS2R38, TAS2R43, TAS2R46) has been identified across various tumor tissues, such as gastric cancer, pancreatic cancer and head and neck squamous cell carcinoma (HNSCC) (Sternini & Rozengurt, 2025). Activating TAS2Rs can produce significant anti-cancer effects, capitalizing on the vulnerability of cancer cell mitochondria and calcium homeostasis. For example, in HNSCC, TAS2R activation leads to mitochondrial depolarization and caspase cascades, which culminate in tumor cell apoptosis (Fig. 3D). Furthermore, high TAS2R expression correlates with improved patient survival outcomes (Carey et al., 2022). In acute myeloid leukemia, bitter agonists such as denatonium benzoate induce cell cycle arrest at the G0/G1 phase and inhibit oxidative phosphorylation (Salvestrini et al., 2020). In breast cancer, TAS2R4/14 can suppress cell migration and proliferation via the MAPK/ERK and G α 12/13-RhoGTPase signaling pathways (Singh et al., 2020). Furthermore, TAS2Rs can reverse drug resistance, remodel the tumor microenvironment and regulate cell fate. In pancreatic ductal adenocarcinoma, the activation of TAS2R10 (e.g., with caffeine) inhibits Akt phosphorylation and downregulates the expression of the multidrug resistance protein ABCG2, thereby significantly enhancing the tumor's sensitivity to chemotherapeutics like gemcitabine (Stern et al., 2018). Concurrently, leveraging the specific high expression of TAS2R9 in cancer-associated fibroblasts, researchers have successfully engineered delivery systems targeting the tumor stroma (Hung et al., 2023). Additionally, the activation of TAS2R8 and TAS2R10 effectively downregulates stemness markers (e.g., CD133, Notch, Sox2) and hypoxia-inducible factor-1 α (HIF-1 α) signaling in neuroblastoma, attenuating their invasive capacity and inducing benign cellular differentiation (Seo et al., 2017).

It is noteworthy that, compared to cancer cells, normal cells possess robust calcium buffering and homeostatic regulatory mechanisms. Therefore, TAS2R activation in normal cells typically elicits only functional modulation (e.g., airway smooth muscle relaxation, intestinal hormone secretion or transient growth arrest/differentiation regulation in hair follicles) rather than cell death (Chen et al., 2025). This differential response underscores their potential as safe therapeutic targets, providing a compelling rationale for developing precise therapies involving TAS1R antagonists (to curb malignant proliferation) and TAS2R agonists (to induce tumor cell apoptosis) without causing collateral damage to normal tissues.

3.3.5. Others

Taste receptors also play crucial roles in the cardiovascular, reproductive and urinary systems. In the cardiovascular system, TAS1Rs and TAS2Rs mediate the regulation of myocardial contraction and relaxation, respectively (Wei et al., 2025). Furthermore, genetic polymorphisms (e.g., loss-of-function variants in TAS2R14, TAS2R30, TAS2R31, TAS2R46 and TAS2R50, as well as gain-of-function variants in TAS2R10) can significantly alter signal responsiveness (Chen et al., 2025). TAS2Rs activation induces vascular smooth muscle relaxation, while TAS1R3 exerts vasculoprotective effects within the renal and retinal microcirculation. Additionally, TAS1R1, TAS1R3 and TAS2Rs have been identified in the testes and spermatozoa, where they are involved in regulating sperm chemotaxis and motility. Notably, TAS1R3

also promotes testosterone synthesis via the cAMP-PKA-SP1 signaling pathway (Liu, Wang, et al., 2025). TAS2Rs are similarly expressed in human ovarian granulosa cells, cumulus cells and the myometrium; their activation by bitter agonists such as chloroquine or phenanthroline exerts a potent relaxant effect on pre-contracted uterine tissues (Zheng et al., 2017). Similarly, TAS2Rs in the bladder contribute to detrusor muscle relaxation and the maintenance of glomerular structure (Zhai et al., 2016). These findings further underscore the broad regulatory roles of taste receptors in maintaining systemic physiological homeostasis.

3.4. Factors influencing the expression and function of ectopic taste receptors

The expression and functions of ectopic taste receptors are regulated by multiple factors: (i) Tissue specificity. For instance, TAS1R1 and TAS1R3 are predominantly expressed in the small intestine and colon, whereas TAS1R2 is primarily localized to the small intestine. Specifically, TAS2R1, TAS2R42, TAS2R45, TAS2R46, TAS2R50 and TAS2R60 are exclusively expressed in the colon, while TAS2R7 and TAS2R8 are confined to the jejunum (Chen et al., 2025). Distinct from the majority of ectopic receptors, TAS2R14 exhibits higher expression levels in the small intestine and pancreas than in the oral cavity, suggesting its functional roles are primarily mediated within extra-oral tissues (Sternini & Rozengurt, 2025). (ii) Dietary patterns and metabolic status. High-fat diets have been shown to upregulate the expression of intestinal TAS2Rs. Obese individuals exhibit an increased population of TAS2R38-positive cells in the colon yet accompanied by a decrease in receptor sensitivity (Liszt et al., 2022). Receptor activation exhibits both ligand specificity and temporal dynamics. For example, short-term exposure to bitter flavanols stimulates ghrelin secretion, whereas chronic administration suppresses the release of this hormone (Trius-Soler & Moreno, 2024). (iii) Genetic polymorphisms and sex differences. Polymorphisms in the TAS2R38 gene not only dictate bitter taste sensitivity but also correlate significantly with body mass index (particularly in females) as well as the efficacy of immune defense against bacterial quorum-sensing molecules (Trius-Soler & Moreno, 2024). Females generally exhibit greater sensitivity to bitter agonist-induced modulation of gastric motility and satiety, implying that sex hormones may participate in regulating taste receptor function (Deloose et al., 2017). Functioning as sophisticated physiological sensors, ectopic taste receptors are collectively modulated by tissue distribution, dietary metabolism and genetic background, constituting a highly coordinated regulatory network that bridges molecular recognition with systemic homeostasis.

4. Translational applications targeting taste receptors: Nutritional and health interventions

4.1. Genotype-guided precision nutrition and personalized dietary interventions

Research indicates that single nucleotide polymorphisms in taste receptor genes are key genetic factors driving individual variations in dietary preferences and intake behaviors (Nolden & Feeney, 2020). For instance, TAS2R38 gene variants significantly modulate individual sensitivity to bitter compounds such as glucosinolates. This genetically predisposed sensory barrier often leads to the long-term avoidance of cruciferous vegetables, which are rich in dietary fiber and antioxidants (Ma & Lu, 2023). Since the phytochemicals in these vegetables play crucial roles in scavenging free radicals, regulating lipid metabolism and maintaining intestinal barrier function, their long-term deficient intake can compromise the body's defense mechanisms, thereby increasing long-term susceptibility to cardiovascular diseases, chronic gastrointestinal disorders and cancer via cumulative metabolic effects (Sharma et al., 2025). Similarly, variants in various bitter taste receptor genes

modulate individual sensitivity to compounds such as epicatechin, tannins, anthocyanins and ethanol, further dictating the dietary proportion and biological impact of chemopreventive phytochemicals or alcohol in an individual's diet (Nolden & Feeney, 2020). Furthermore, genetic variations in TAS1R2/TAS1R3 significantly affect sensitivity to and craving for free sugars. Individuals with a high-threshold phenotype usually consume more sweets to achieve equivalent sensory reward, serving as a key risk factor for obesity, type 2 diabetes and early-onset dental caries (Eny et al., 2010). A comprehensive understanding of taste receptor genetic variations enables precision nutrition to evolve from generalized dietary guidelines to deeply customized interventions based on individual "genetic-sensory profiles". This profiling allows for the prediction of specific nutrient intake biases, leveraging genetic predispositions to guide precise behavioral modifications, thereby preventing diet-related diseases and improving overall quality of life (Prakrithi et al., 2022). Conversely, dietary intake and exposure also dynamically influence taste receptor expression (Nolden & Feeney, 2020). Ectopic taste receptors provide a multidimensional regulatory pathway for precision nutrition, spanning from "sensory drive" to "metabolic feedback". Their genetic polymorphisms further explain why individuals exhibit divergent metabolic responses to identical diets (Lee et al., 2012). Future precision nutrition strategies should integrate genomics, sensomics and nutritional metabolomics to construct multimodal predictive models. These models will not only enable precise meal formulation based on sensory preferences but also optimize intervention timing and dosage according to the "sensing characteristics" of the individual's metabolic system, thereby realizing a true "sensory-metabolic-health" personalized management pathway.

4.2. Taste receptors as non-invasive diagnostic and prognostic biomarkers for metabolic and malignant diseases

The widespread expression of taste receptors across the human body, coupled with their regulatory roles in core metabolic pathways, endows them with substantial potential as biomarkers for disease diagnosis and prognosis. For instance, TAS2R14 is commonly downregulated in various tumors, including thyroid cancer, and its expression levels may serve as a potential diagnostic indicator of malignancy. Moreover, the differential expression of TAS2R14 among distinct pathological subtypes of thyroid cancer (e.g., tall papillary thyroid cancer subtype vs. classical and follicular papillary thyroid cancer subtypes) highlights its utility in assisting cancer molecular subtyping (Duarte et al., 2020). Notably, patients with chronic rhinosinusitis exhibit decreased sensitivity to the bitter compound denatonium and increased sensitivity to sucrose (Seah et al., 2023). This opens novel avenues for non-invasive functional diagnostics, demonstrating the broad prospects of taste receptors in disease detection. Furthermore, taste receptors can act as prognostic biomarkers in multiple cancers, including breast, prostate, head and neck squamous cell carcinomas (Yasin et al., 2024), potentially guiding the selection of appropriate cancer treatment intensities (Carey et al., 2022). The expression levels of various taste receptors correlate strongly with patient survival outcomes and exhibit high tissue specificity. For example, elevated TAS1R1 expression is associated with improved survival in lung adenocarcinoma and non-papillary bladder cancer, while increased TAS2R14 expression correlates with worsened survival in adrenocortical carcinoma and esophageal adenocarcinoma, as well as lower survival rates in thyroid cancer patients (Carey, Kim, et al., 2022). In tumors such as clear cell renal cell carcinoma, multiple TAS2R genes collectively form prognostic gene signatures, suggesting potential synergistic effects. This prognostic prediction capability extends to chronic rhinosinusitis as detailed in Section 3.2.2. Patients homozygous for the AVI TAS2R38 polymorphism typically exhibit poorer postoperative outcomes, requiring more interventions (Seah et al., 2023). Despite their immense potential, translating taste receptors into clinically actionable diagnostic and prognostic biomarkers remains challenging: (i) Their expression and function are highly tissue-

and disease-specific; the same receptor (e.g., TAS2R14) may exhibit opposing survival associations in different cancers, necessitating rigorous disease subtyping for any clinical application. (ii) The endogenous ligands of taste receptors and the specific downstream signaling pathways they drive during disease progression remain unclear, leading to a lack of depth mechanistic interpretation for these biomarkers. (iii) Most current studies lack large-scale, prospective clinical trials to validate their specificity, sensitivity and practical clinical utility (Carey, Kim, et al., 2022).

4.3. Metabolic regulation and drug discovery targeting taste receptors

Currently, approximately 34% of FDA-approved drugs target GPCRs (Gu et al., 2025). As critical members of the GPCR family, ectopic taste receptors play pivotal roles in regulating key metabolic processes, including gastrointestinal hormone secretion and insulin release. Notably, at least 15 FDA-approved clinical drugs have already been identified to activate TAS2R14 (Levit et al., 2014). Therefore, the in-depth development of targeted therapeutics directed at taste receptors has emerged as a crucial strategic direction for combating complex diseases like type 2 diabetes and obesity. Various natural and synthetic compounds targeting taste receptors (such as KDT501, a TAS2R1 agonist for diabetes treatment) have advanced to clinical trials (Yang et al., 2025). Chou (2021) proposed that specific agonists targeting bitter taste receptors (e.g., human TAS2R1 and TAS2R38) may yield metabolic benefits and anti-inflammatory effects by modulating intestinal hormone secretion and bile acid metabolism in patients with metabolic syndrome or diabetes complicated by dyslipidemia. Furthermore, studies have also shown that the combination of bitter receptor agonists (e.g., KDT501) with DPP-IV inhibitors (e.g., sitagliptin) synergistically improves glucose tolerance in diet-induced obese mice (Kok et al., 2018). Additionally, various TAS2R agonists, such as parthenolide, quinine, xanthohumol, apigenin, andrographolide and tangerine peel extract, have demonstrated potential for lowering blood glucose, reducing body weight and reversing airway remodeling (Li et al., 2025; Trius-Soler & Moreno, 2024; Yang et al., 2025). The multi-target actions of natural compounds offer broader possibilities for their application in treating metabolic diseases. Specifically, structural optimization can further enhance their pharmacological properties. For instance, DMAPT, a derivative obtained through the structural modification of parthenolide, not only exhibits improved water solubility and bioavailability but also effectively induces apoptosis in leukemia cells (Li et al., 2025). Furthermore, researchers have utilized nanocarrier encapsulation for bitter compounds to mask unpalatable tastes and enhance bioavailability (Vishvakarma et al., 2023). The development of fast-dissolving oral films allows for rapid drug dissolution, maximizing the efficacy of active pharmaceutical ingredients, particularly beneficial for children or the elderly with chewing or swallowing difficulties (Saxena & Singh, 2022). These advancements provide critical references for developing taste receptor-targeted drugs with greater clinical viability. However, a major bottleneck in current translational research is the lack of rigorous target validation. While numerous studies report promising health benefits in cellular or animal models, they often rely on broad-spectrum ligands (e.g., quinine or denatonium). Given the vast number of taste receptor family members, their prevalent broad-spectrum ligand recognition characteristics and the historical absence of reliable structural templates, drug design is highly susceptible to poor selectivity and potential off-target effects (Löber & Gmeiner, 2022). It is worth noting that, in certain therapeutic contexts, non-selective (promiscuous) TAS2R agonists or antagonists may offer advantages by enabling multi-target synergistic effects (Hu et al., 2024). However, the broad-spectrum binding properties of such non-selective ligands inevitably raise concerns about off-target toxicity. Theoretically, modulators that are promiscuous among multiple TAS2R subtypes but do not affect other class A GPCRs are feasible, given the substantial structural and sequence divergence between TAS2Rs and class A GPCRs. Nevertheless, achieving

high selectivity among different TAS2R subtypes within the same family is equally challenging, because TAS2R subtypes exhibit considerable structural conservation within their binding pockets. This intra-family similarity makes it difficult to distinguish individual TAS2R subtypes and to achieve subtype-selective modulation. Consequently, the scarcity of highly selective pharmacological tools—particularly selective antagonists—makes it exceedingly difficult to pinpoint the exact receptor subtype responsible for a specific physiological outcome. Indeed, the development of selective antagonists has traditionally lagged, yet they are absolutely critical for establishing receptor-specific effects *in vivo* and enabling definitive loss-of-function studies. Furthermore, it is crucial to note that while traditional high-throughput screening and medicinal chemistry have yielded numerous modulators for classic GPCRs over the past decades (Yasi et al., 2020), taste receptors present unique obstacles that render these classical approaches exceptionally challenging. Unlike canonical GPCRs that bind endogenous ligands with nanomolar affinity, taste receptors naturally exhibit low-affinity (micromolar) and broad-spectrum ligand recognition (Di Pizio et al., 2016). This leads to poor signal-to-noise ratios and high false-positive or false-negative rates in high-throughput screening assays (Tan & Seetoh, 2022). Taste receptors are also notoriously difficult to heterologously express and traffic to the plasma membrane in native-like conformations. Consequently, empirical screening frequently fails to identify compounds targeting the highly divergent and non-canonical binding interfaces of taste receptors. In addition, the high structural conservation within the TAS2R family further complicates the design of selective ligands targeting individual subtypes, an issue that is often less prominent in classical GPCR drug discovery. This intra-family similarity underscores the need for strategies that can distinguish subtle differences among TAS2R binding pockets. Encouragingly, the integration of newly resolved cryo-EM structures (such as TAS2R14, TAS2R43 and TAS2R46) with advanced AI predictive models offers a powerful solution. Researchers can now conduct ultra-large-scale *in silico* screening of vast chemical libraries to develop “tailored” ligands for specific receptor conformations. For example, utilizing structure-based virtual screening combined with molecular dynamics, investigators can identify compounds that selectively bind to unique topological features—such as the distinct extended TM5 of TAS2R43 or the intracellular allosteric pocket of TAS2R14—thereby avoiding cross-reactivity with the other 25 TAS2R members. Beyond structure-based virtual screening, *in silico* approaches can be further leveraged to directly address promiscuity and off-target risks. For example, reverse docking can evaluate the activity of candidate compounds against multiple TAS2R subtypes and class A GPCRs in parallel, thereby assessing their potential off-target effects on other GPCR families. Machine learning models trained on known TAS2R ligands can predict selectivity profiles across the entire TAS2R repertoire. When combined with structural comparison and molecular dynamics simulations, it becomes possible to design screening motifs that recognize TAS2R-specific structural features, ultimately identifying safe modulators that are promiscuous within the TAS2R family but devoid of activity on class A GPCRs. Targeting these distinct pockets offers opportunities for rationally designing selective PAMs or antagonists that do not compete with natural orthosteric agonists. Developing these highly selective agonists and antagonists via AI-driven pipelines drives a fundamental transition from “blind screening” to “rational design”. When coupled with functional testing in receptor-knockout translational models, this transition facilitates rigorous target validation *in vivo* and represents a critical stepping stone toward future viable clinical therapeutics. Furthermore, many promising natural agonists still suffer from pharmacokinetic deficiencies, such as low oral bioavailability and poor chemical stability. Species differences also hinder the effective translation of preclinical animal data into clinical practice. Therefore, striking a balance between high drug efficacy and specificity, while overcoming the delivery bottlenecks associated with peripheral administration, remains the core challenge urgently needing resolution in the current development of taste receptor-targeted therapeutics.

Crucially, although a vast body of literature indicates that numerous taste receptors exert beneficial effects across various diseases, simultaneously developing therapeutics against all 26 subtypes is neither clinically feasible nor efficient. Moreover, the inherent heterogeneity in experimental designs across existing studies precludes effective cross-comparisons to definitively determine which specific receptor yields the greatest therapeutic efficacy. To effectively guide clinical translation, it is imperative to establish a strategic target prioritization framework. Herein, based on the current functional consensus (as described in Section 3.3), the availability of experimental structural templates (as described in Section 2.2) and the fact that certain FDA-approved clinical drugs function as specific bitter taste receptor activators, we propose several potentially viable candidate targets for specific clinical indications. For metabolic regulation, TAS2R14, TAS2R43 and TAS2R46 represent promising primary candidates due to their resolved structures. Particularly, TAS2R14 is among the most extensively documented subtypes, showing a robust capacity to enhance GLP-1 secretion and improve glucose homeostasis. Following a similar rationale, TAS2R43 may emerge as a highly suitable target to combat low-grade intestinal inflammation (leaky gut). In the context of respiratory and oral health, both TAS2R14 and TAS2R38 have the potential to serve as core therapeutic hubs for asthma and gingival diseases. Additionally, TAS2R14 demonstrates considerable translational promise in oncology and central nervous system disorders, whereas TAS2R38 could also be considered a principal candidate for maintaining dermatological homeostasis. Concentrating drug discovery pipelines on these structurally informed and functionally validated targets is expected to minimize off-target liabilities and could significantly accelerate the translation from bench to bedside.

Further evaluation of these opportunities through the lenses of developmental feasibility and clinical impact allows for a clear distinction between “low-hanging fruits” and “high-impact breakthroughs”, thereby accelerating the translational cycle of taste receptor research. On one hand, nutraceutical discovery and localized, barrier-targeted interventions represent the “low-hanging fruits” of taste receptor pharmacology. Utilizing dietary phytochemicals (e.g., resveratrol) as functional food ingredients or nutraceuticals for metabolic conditioning offers an accelerated, commercially viable translational pathway due to their established safety profiles and flexible regulatory hurdles. Similarly, localized formulations—such as topical creams for dermatological inflammation (targeting TAS2R38), nasal sprays for chronic rhinosinusitis or oral rinses for gingival diseases (targeting TAS2R14/38)—bypass systemic pharmacokinetic bottlenecks (e.g., low oral bioavailability) and minimize systemic off-target toxicities by restricting receptor modulation to peripheral tissue barriers. On the other hand, systemic small-molecule modulators targeting global chronic diseases offer the “high-impact breakthroughs”, albeit with steeper developmental hurdles. Drugging the structurally elucidated TAS2R14/43/46 axis to robustly drive endogenous GLP-1 co-secretion could fundamentally disrupt obesity and type 2 diabetes therapeutic landscape. Likewise, engineering highly selective, desensitization-resistant TAS2R14 allosteric agonists as next-generation bronchodilators, alongside TAS2R43-targeted agents to fortify the intestinal barrier, holds immense promise to revolutionize the management of severe asthma and leaky gut syndrome. Balancing these immediate, peripheral nutraceutical avenues with high-risk, high-reward systemic pharmaceutical strategies will ultimately unleash the full translational potential of taste receptor-targeted therapeutics.

5. Conclusion and future perspectives

This review provides a systematic overview of the recent advances in the structure, function, and nutritional and health interventions of human sweet, umami and bitter taste GPCRs. In recent years, the synergistic development of cryo-EM and AI (e.g., AlphaFold) has successfully resolved the high-resolution three-dimensional structures of

TAS2R14, TAS2R46 and the TAS1R2/TAS1R3 heterodimer, along with their unique ligand recognition and activation mechanisms. This breakthrough fills a long-standing gap in the structural biology of taste receptors and establishes a solid structural foundation for understanding their functions. Mounting evidence reveals that taste receptors are sophisticated chemical sensors that extend far beyond the oral cavity, being widely distributed throughout the body. By recognizing dietary components, microbial metabolites and endogenous signaling molecules, they initiate highly conserved intracellular signaling cascades, playing pivotal roles in core physiological processes such as regulating taste perception, maintaining metabolic homeostasis, coordinating immune responses and controlling cell fate. These functions not only influence individual dietary preferences and behaviors but are also deeply associated with the susceptibility and prognosis of complex diseases such as obesity, diabetes, asthma and cancer. Therefore, taste receptors have emerged as crucial potential targets for precision nutrition interventions, disease diagnosis and prognostic assessment, and novel drug discovery.

Despite significant progress in our understanding of taste receptors, this field still faces formidable challenges and offers vast avenues for exploration. Currently, the high-resolution structures of the TAS1R1/TAS1R3 heterodimer and the majority of TAS2R subtypes remain unresolved. Future research urgently needs to overcome the technical bottlenecks of heterologous expression systems. By optimizing cell line construction and leveraging cutting-edge protein engineering technologies (e.g., Nanodiscs or high-affinity antibody scaffolds), stable and high-level expression of taste receptor proteins can be achieved, particularly resolving the *in vitro* co-expression and correct folding bottlenecks of the TAS1R1/TAS1R3 heterodimer. These collective technological breakthroughs will enable cryo-EM to capture the dynamic conformational changes of these receptors across distinct functional states, providing a “gold standard” for the kinetic analysis of ligand recognition and signal transduction. Furthermore, utilizing the resolved high-resolution structures of taste receptors, combined with AI-driven virtual screening and *de novo* design, can accelerate the large-scale discovery of novel exogenous ligands and potential drug candidates. Nonetheless, translating these computational predictions into clinically relevant therapeutics still requires rigorous *in vivo* validation. Given the substantial differences in receptor orthologs between rodents and humans, humanized mouse models remain indispensable at the current stage. Computational screening can help narrow down candidate compounds targeting human TAS2Rs and reduce the burden on animal testing. In this complementary framework, *in silico* methods prioritize and refine leads, while humanized mouse models provide the complex systemic and signaling context that computation alone cannot replicate, offering essential *in vivo* evidence for successful translation. Simultaneously, integrating multi-omics and computational biology approaches is essential to systematically reveal the endogenous ligand networks of taste receptors. Employing techniques such as cell-specific knockouts, organoids, real-time imaging and spatial omics will allow for the precise dissection of the dynamic response mechanisms of taste receptors within specific tissue microenvironments. These will elucidate their precise regulation of systemic metabolism and immunity, thereby laying a robust theoretical foundation for developing more targeted precision nutrition interventions and innovative drugs with minimized side effects. To effectively translate taste receptor research into practical nutritional and health interventions, there is a pressing need to promote large-scale clinical validation and standardization of taste receptors as diagnostic and prognostic biomarkers, and further to develop novel non-invasive diagnostic strategies. Building upon the structure-based rational design of highly selective modulators or biased ligands, concerted efforts should be made to optimize targeted delivery systems to enhance their chemical stability and therapeutic efficacy. Additionally, exploring the combinatorial application of taste receptor-targeted drugs with existing therapies (such as GLP-1 receptor agonists) will provide safer and more efficient synergistic intervention strategies for

complex chronic diseases like metabolic syndrome. As multidisciplinary technologies continue to converge and deepen, these systemic “chemical sentinels” will undoubtedly provide a novel perspective on human health and disease, as well as bring groundbreaking innovative solutions to personalized nutrition and health interventions.

CRedit authorship contribution statement

Yuxiang Gu: Conceptualization, Writing - original draft. Yixuan Cui: Writing - review & editing. Kangli Guo: Writing - review & editing. Guoping Zhao: Writing - review & editing. Xing Li: Writing - review & editing. Weixi Yang: Writing - review & editing. Yanbo Wang: Writing - review & editing.

Conflicts of interest

There are no conflicts of interest to declare.

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Data availability

No data was used for the research described in the article.

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